

## MARSHFIELD, WI, USA

WMO: 726574

Lat: 44.638N

Lon: 90.188W

Elev: 1255

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94985

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.7      | -8.1 | -20.3                         | 1.9 | -11.7 | -15.6 | 2.5 | -7.2 | 25.6                  | 13.0 | 23.2 | 14.7 | 8.0                   | 280  | 0.584 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.1                   | 87.5            | 72.8 | 84.1 | 70.7 | 81.5 | 69.2 | 75.1                | 83.7 | 73.3 | 81.4 | 71.4 | 78.8 | 9.9                  | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.4                            | 125.8 | 80.2 | 70.4 | 117.4 | 78.2 | 68.8 | 110.8 | 76.6 | 39.5          | 83.9 | 37.6 | 81.7 | 36.0 | 79.0 | 83.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 23.1          | 19.9 | 18.0 | DB                         | -18.9 | 91.9               | 6.3 | 4.1                                                | -23.4 | 94.8       | -27.1 | 97.2       | -30.6 | 99.5       | -35.2 | 102.5 | (4) |
| (5)               |      |      | WB                         | -19.2 | 78.5               | 6.2 | 2.4                                                | -23.7 | 80.2       | -27.3 | 81.6       | -30.8 | 82.9       | -35.3 | 84.6  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.3        | 14.7  | 18.7  | 30.8  | 44.0  | 56.1  | 65.5  | 69.8  | 67.6  | 60.4  | 47.3  | 33.6  | 21.3  |  |  |
|                                                                        | DBStd       | 21.40       | 12.62 | 12.11 | 12.11 | 9.61  | 8.32  | 6.75  | 5.58  | 5.56  | 7.99  | 9.00  | 10.27 | 11.38 |  |  |
|                                                                        | HDD50       | 4395        | 1095  | 876   | 605   | 224   | 34    | 1     | 0     | 0     | 11    | 163   | 498   | 888   |  |  |
|                                                                        | HDD65       | 7997        | 1560  | 1296  | 1060  | 630   | 297   | 75    | 18    | 35    | 180   | 552   | 941   | 1353  |  |  |
|                                                                        | CDD50       | 2314        | 0     | 0     | 11    | 46    | 222   | 464   | 615   | 546   | 324   | 79    | 7     | 0     |  |  |
|                                                                        | CDD65       | 441         | 0     | 0     | 0     | 1     | 20    | 89    | 168   | 116   | 44    | 3     | 0     | 0     |  |  |
|                                                                        | CDH74       | 3702        | 0     | 0     | 2     | 34    | 225   | 744   | 1429  | 890   | 349   | 29    | 0     | 0     |  |  |
|                                                                        | CDH80       | 935         | 0     | 0     | 0     | 5     | 50    | 187   | 411   | 207   | 73    | 2     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 8.4         | 9.1   | 9.1   | 9.0   | 9.9   | 9.1   | 7.6   | 7.0   | 6.6   | 7.4   | 8.6   | 9.0   | 8.9   |  |  |
|                                                                        | PrecAvg     | 33.5        | 1.0   | 0.9   | 1.9   | 3.0   | 4.3   | 4.3   | 3.8   | 4.1   | 4.3   | 2.6   | 2.1   | 1.3   |  |  |
| (15) Precipitation                                                     | PrecMax     | 45.9        | 2.7   | 3.0   | 4.7   | 5.7   | 10.4  | 9.6   | 10.5  | 9.6   | 10.6  | 5.6   | 5.7   | 3.7   |  |  |
|                                                                        | PrecMin     | 21.6        | 0.1   | 0.0   | 0.1   | 0.5   | 1.0   | 0.7   | 0.5   | 0.7   | 0.3   | 0.3   | 0.0   | 0.3   |  |  |
|                                                                        | PrecStd     | 5.4         | 0.6   | 0.7   | 1.1   | 1.4   | 2.2   | 2.4   | 2.1   | 2.0   | 2.5   | 1.3   | 1.3   | 0.8   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 44.2  | 49.6  | 70.0  | 79.1  | 85.5  | 90.3  | 92.7  | 89.7  | 86.5  | 78.6  | 64.3  | 52.1  |  |  |
|                                                                        |             | MCWB        | 38.1  | 43.0  | 59.5  | 61.2  | 68.4  | 74.9  | 76.4  | 74.4  | 71.2  | 65.2  | 53.2  | 49.2  |  |  |
|                                                                        | 2%          | DB          | 37.8  | 43.0  | 60.7  | 70.8  | 80.3  | 84.5  | 87.9  | 85.2  | 81.6  | 72.1  | 57.4  | 43.3  |  |  |
|                                                                        |             | MCWB        | 34.5  | 38.1  | 51.7  | 55.4  | 65.0  | 70.7  | 73.8  | 72.0  | 68.6  | 61.6  | 50.5  | 38.9  |  |  |
|                                                                        | 5%          | DB          | 34.3  | 38.2  | 53.8  | 65.3  | 75.2  | 81.5  | 84.1  | 81.7  | 78.5  | 66.5  | 53.1  | 38.6  |  |  |
|                                                                        |             | MCWB        | 32.6  | 34.8  | 47.2  | 52.1  | 61.9  | 69.0  | 71.6  | 69.8  | 66.7  | 58.1  | 46.5  | 35.6  |  |  |
|                                                                        | 10%         | DB          | 32.0  | 35.0  | 47.3  | 60.6  | 71.4  | 78.6  | 81.5  | 79.1  | 74.1  | 62.2  | 48.2  | 35.5  |  |  |
|                                                                        |             | MCWB        | 30.4  | 32.4  | 41.4  | 48.9  | 59.3  | 66.8  | 69.8  | 67.7  | 64.5  | 55.1  | 43.2  | 33.4  |  |  |
|                                                                        | 0.4%        | WB          | 38.5  | 43.8  | 60.1  | 63.2  | 72.0  | 76.2  | 79.1  | 77.0  | 73.3  | 69.1  | 56.5  | 49.0  |  |  |
|                                                                        |             | MCDB        | 42.4  | 49.1  | 68.3  | 74.8  | 82.1  | 85.7  | 89.6  | 84.9  | 82.3  | 73.5  | 61.3  | 51.5  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 35.1  | 38.3  | 53.9  | 57.9  | 67.6  | 73.5  | 75.8  | 74.1  | 70.9  | 63.3  | 51.8  | 40.0  |  |  |
|                                                                        |             | MCDB        | 37.7  | 42.1  | 58.5  | 68.7  | 76.3  | 82.3  | 84.5  | 82.1  | 78.7  | 68.7  | 57.1  | 42.3  |  |  |
|                                                                        | 5%          | WB          | 32.8  | 35.3  | 46.9  | 53.8  | 64.3  | 71.1  | 73.7  | 72.0  | 68.7  | 59.7  | 46.8  | 35.7  |  |  |
|                                                                        |             | MCDB        | 34.3  | 38.2  | 53.7  | 61.6  | 72.3  | 79.0  | 81.6  | 79.2  | 75.1  | 65.8  | 51.4  | 38.2  |  |  |
|                                                                        | 10%         | WB          | 30.3  | 32.3  | 41.7  | 50.0  | 61.4  | 68.5  | 71.6  | 70.0  | 66.2  | 55.5  | 43.4  | 33.1  |  |  |
|                                                                        |             | MCDB        | 31.8  | 34.7  | 47.2  | 59.7  | 68.6  | 75.9  | 78.9  | 76.5  | 72.7  | 61.4  | 48.0  | 34.9  |  |  |
|                                                                        | 5% DB       | MDBR        | 14.9  | 16.0  | 17.1  | 19.8  | 20.4  | 19.4  | 20.1  | 19.8  | 20.2  | 18.0  | 14.6  | 13.2  |  |  |
|                                                                        |             | MCDBR       | 16.1  | 17.5  | 23.6  | 28.9  | 25.3  | 22.9  | 22.2  | 22.1  | 24.0  | 24.0  | 21.9  | 16.7  |  |  |
|                                                                        |             | MCWBR       | 13.7  | 13.6  | 15.4  | 16.7  | 13.0  | 12.0  | 10.8  | 10.9  | 12.4  | 14.7  | 15.1  | 13.5  |  |  |
|                                                                        |             | MCWBR       | 15.5  | 16.3  | 22.0  | 25.3  | 22.5  | 20.5  | 19.9  | 19.6  | 20.4  | 21.4  | 20.0  | 14.9  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.248 | 0.273 | 0.311 | 0.346 | 0.379 | 0.380 | 0.374 | 0.378 | 0.349 | 0.316 | 0.285 | 0.257 |  |  |
|                                                                        |             | taud        | 2.409 | 2.335 | 2.387 | 2.379 | 2.330 | 2.355 | 2.382 | 2.369 | 2.452 | 2.529 | 2.529 | 2.456 |  |  |
|                                                                        | Ebn at Noon |             | 283   | 292   | 294   | 291   | 282   | 282   | 277   | 278   | 274   | 267   | 267   | 268   |  |  |
|                                                                        |             | Edh at Noon | 23    | 30    | 33    | 36    | 39    | 38    | 37    | 36    | 30    | 25    | 21    | 20    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 543   | 901   | 1252  | 1436  | 1711  | 1877  | 1944  | 1648  | 1302  | 824   | 522   | 391   |  |  |
|                                                                        | RadStd      |             | 61    | 83    | 140   | 144   | 132   | 146   | 94    | 110   | 90    | 96    | 51    | 47    |  |  |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)                          | (b) | (c)   | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            |     |       | N/A     | -2.70  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (44 neighbors) |     |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page